

White-light frequency comb generation with a diode-pumped Cr:LiSAF laser

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Received January 3, 2001

We have created a broad spectrum spanning more than an optical octave by launching femtosecond pulses from a battery operated Cr:LiSAF laser into a photonic crystal fiber. Despite the massive broadening in the fiber, the comb structure of the spectrum is preserved, and this frequency comb is perfectly suited for applications in optical frequency metrology. © 2001 Optical Society of America

OCIS codes: 120.3930, 320.7090.

Spectral broadening of femtosecond (fs) laser pulses in specially designed so-called photonic crystal fibers^{1–4} (PCFs) has recently become a powerful tool to deliver broad spectra with precise frequency combs for frequency metrology^{5,6} or as bright white-light sources in applications such as optical coherence tomography. So far, such experiments have used Kerr-lens (KL) mode-locked (ML) Ti:sapphire lasers. Although these lasers are convenient laboratory workhorses, they cannot be directly diode pumped; instead, costly large-frame Ar-ion or frequency-doubled Nd:YAG lasers are needed for pumping.

In this Letter we report results obtained with a directly diode-pumped battery operated Cr:LiSAF laser to achieve a spectrum that spans an optical octave. We have successfully investigated the possibility of using this spectrum in optical frequency metrology.

The relatively new laser crystals of the colquirites such as Cr³⁺LiSAF, Cr³⁺LiSGaF, and Cr³⁺LiCaF, which can be directly pumped by laser diodes at 670 nm, offer low-cost high-performance alternatives to Ti:sapphire. These crystals have only a slightly smaller gain bandwidth and even a 2-times-lower saturation intensity than Ti:sapphire. However, there are some disadvantages: The colquirites are soft, their thermal conductivity is more than 10 times lower than in sapphire, at a crystal temperature above 60 °C thermal quenching substantially lowers the gain, and the nonlinearity (needed for KL mode locking) is 5–10 times less than in sapphire. To get high gain and high nonlinearity in a standard Z-fold cavity one has to focus tightly into the laser crystal. The laser diodes in the required power range have a rather poor beam quality, with an M^2 of $\sim 2 \times 8$ measured behind the collimating microlens. To maximize the gain the overlap of the pump beam with the resonator mode was numerically optimized for both axes.⁷ To implement the calculated improvements we used the

setup in Fig. 1. The plane folding mirrors between the crystal and the curved mirrors and microlenses in front of the laser diodes allow for almost aberration-free imaging into the laser crystal and a compact setup. For dispersion compensation one of the highly reflective mirrors was replaced by a chirped mirror. All mirrors in the setup were custom designed and coated by Layertec GmbH (Mellingen, Germany). Because of the low power level of 2×350 mW available from the pump diodes, the high intracavity power required for KL mode locking can be achieved only with a low output coupling of 3% and dense low-scatter sputtered mirror coatings with losses of 0.02% per bounce. The optimized pumping scheme and the low losses produced an overall optical slope efficiency of 39%.

Theoretical and experimental data predict best performance of soft-aperture KL ML Ti:sapphire lasers if the laser crystal is moved out of the center between the curved mirrors in such a way that the beam waist is located on one crystal surface.⁸ But our Cr:LiSAF laser experiments and numerical calculations,⁹ which take into account the strong gain saturation, show that the best performance is accomplished with a higher eccentricity (2.5 mm for a 5-mm Cr:LiSAF crystal and a radius of curvature of 100 mm). We obtained as much as 150 mW of ML power at a 93-MHz repetition rate and dispersion-controlled central wavelengths of 835–895 nm. The ML bandwidth varied from 20 to 30 nm FWHM, with corresponding pulse durations of 60 to 40 fs.

In the reported experiments the laser was operated with 115-mW ML power at 93 MHz, 894-nm central wavelength, and 24-nm FWHM bandwidth (Fig. 2, thick curve), supporting a 40-fs pulse width. The pulses launched into the fiber without compression had a 57-fs pulse width and a pulse-bandwidth product of $\tau\Delta\nu = 0.45$. To obtain good long-term

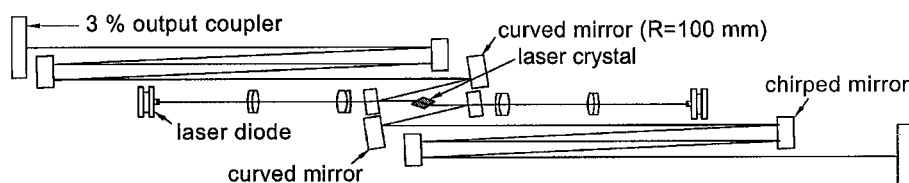


Fig. 1. Laser setup with 26 reflections per round trip to enable the entire apparatus to fit in a $13 \text{ cm} \times 16 \text{ cm} \times 48 \text{ cm}$ box. For details see text.

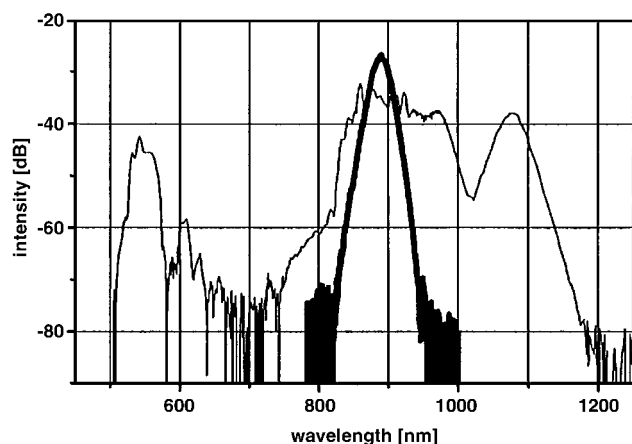


Fig. 2. Spectrum before (thicker curve, middle) and after broadening in the fiber.

stability and low noise, we paid special attention to the mechanical setup. The internal base plate is mechanically, thermally, and electrically isolated from the environment. The completely shielded power supplies are hosted in the same air-tight box, so only 12 V dc ($<1 \text{ A}$) from a battery and water flow for thermal stabilization have to be supplied externally. The use of a single tension-free material ensures the long-term stability of the resonator setup.

Since the introduction in 1998 of fs laser pulses to measure large differences between optical frequencies,¹⁰ this technique is gaining widespread use in many laboratories around the world. The frequency comb of equally spaced modes produced by the pulse train of a fs laser can be conveniently used as a ruler in frequency space. The surprisingly simple basic idea for measuring optical frequencies is to bridge the frequency interval between different harmonics of an optical frequency f with this ruler.¹¹ Femtosecond frequency combs created with Ti:sapphire lasers have been tested thoroughly, and so far no deviation from the regular mode spacing has been discovered.^{6,12}

The broad output spectrum of a fs laser consists of a comb of mode frequencies f_n separated by repetition frequency f_r and with a general offset f_0 (Ref. 11):

$$f_n = nf_r + f_0, \quad n \text{ a large integer.} \quad (1)$$

With known radio frequencies f_r and f_0 , the absolute frequencies of all modes f_n are known. Whereas f_r is readily measurable, f_0 is not easily accessible unless the frequency comb contains more than an optical octave. In this case the infrared part of the spectrum near 1060 nm can be frequency doubled and a beat note with the green part can be observed.^{6,13,14} Now con-

sider a mode in the infrared with frequency $f_1 = n_1 f_r + f_0$. Frequency doubling yields $2 \times f_1 = 2n_1 f_r + 2f_0$. If the frequency comb spans an octave, there will be a mode in the green with mode number $n_2 = 2n_1$ and frequency $f_2 = n_2 f_r + f_0 = 2n_1 f_r + f_0$. Observing a beat signal between the two directly yields $2f_1 - f_2 = f_0$. This comparison results in a $f:2f$ frequency chain that links the radio-frequency regime (f_r, f_0) to the optical domain (comb modes f_n).

To achieve an octave-spanning frequency comb, we spectrally broaden the output spectrum from the fs laser in a PCF fiber. In such a fiber, light is confined to a small solid silica strand suspended in air by thin (100–150-nm-width) threads of silica (see Fig. 3). This fiber is fabricated in a manner similar to that previously reported.¹ The large index contrast between the pure-silica core and the cladding, which is predominantly air, permits the design of fibers with widely different properties from those of conventional fibers. In particular, the strong confinement permits the use of a very small, 1- μm , core diameter, which increases the nonlinear interaction of light with silica. At the same time, the strong waveguide dispersion of the fiber substantially compensates for the material dispersion of silica at wavelengths below 1 μm . As a result, short optical pulses travel further in these fibers before being dispersed, and this further increases the nonlinear interaction. Consequently, broad spectra can be generated in PCFs at relatively low peak powers.^{3,4} The creation of additional spectral modes by the fiber action can be understood as being due either to self-phase modulation or alternatively in the frequency domain to four-wave mixing. However, self-phase modulation makes the pulse susceptible to pump intensity noise, and there are other processes such as stimulated Raman and Brillouin scattering, shock-wave interaction, and modulation instability that might spoil the usefulness of these broadened frequency combs. Nonetheless, a highly phase-coherent pulse train could be generated

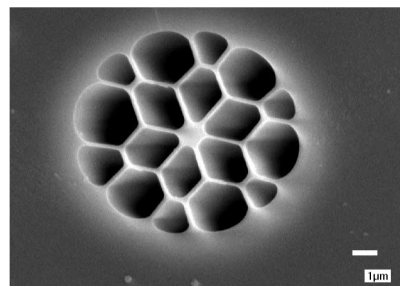


Fig. 3. Core area of the photonic crystal fiber used in the experiment.

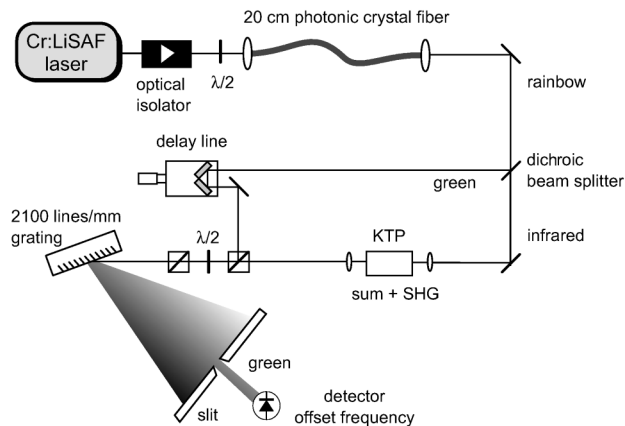


Fig. 4. $f:2f$ optical frequency synthesizer based on stabilization of the interval between an optical frequency f and its second harmonic $2f$ with the help of a fs frequency comb. Once the offset frequency and the repetition rate are locked, all the modes in the comb can be used for optical frequency measurements. SHG, second-harmonic generation.

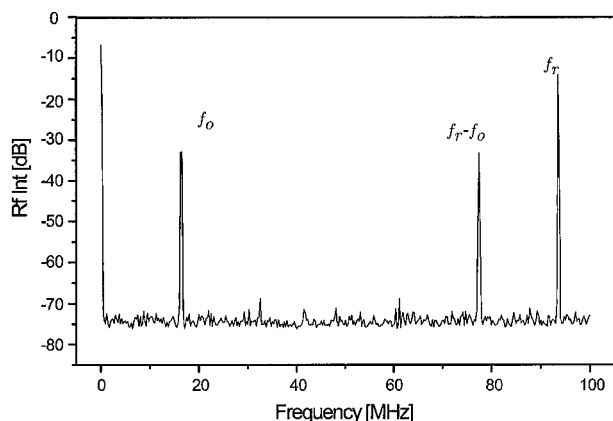


Fig. 5. Radio-frequency spectrum (Rf Int), showing the offset frequency beat. The detection bandwidth was 100 kHz.

with the laser–fiber combination presented here. We have used the PCF with the smallest core diameter available (see Fig. 3, $1\ \mu\text{m}$) with zero group-velocity dispersion near 580 nm to increase the nonlinear interaction. The spectra generated in PCFs with small cores and short group-velocity dispersed zero wavelengths generally cover a broad range but exhibit deep spectral holes, governed by the group-velocity dispersion zero and the pump wavelength. This result is due to the complicated phase structure of the pulse propagation in the fiber, which also has so far prohibited the recompression of such ultrabroad spectra. A pronounced example of a broad spectrum with deep spectral holes is shown in Fig. 2. The peaks near 530 and 1060 nm nevertheless facilitate self-referencing of the frequency comb as described above.

Starting with 115 mW of power from the laser, 20 mW is lost in an optical isolator (Gsänger, DLI

1), and 42 mW average power is coupled through 20 cm of PCF, which results in the spectrum shown in Fig. 2. The nonlinear interferometer shown in Fig. 4 has been described in detail.⁶ Figure 5 shows the resultant rf spectrum at the photodetector in Fig. 4. A beat signal f_0 with a signal-to-noise ratio exceeding 40 dB in a 100-kHz bandwidth has been obtained. The signal-to-noise ratio is sufficient to phase lock both f_0 and f_r . Without any stabilization, the Allan standard deviation of the repetition frequency of the free-running laser at 1 s was 2.2×10^{-10} . We have not stabilized the comb because of the limited access to the sealed box of the LiSAF laser. With the addition of one more piezoelectric transducer to control the cavity length and direct control of the pump power, both parameters will be controlled in a future frequency chain application, which will result in a compact and transportable optical frequency synthesizer without the need for large-frame pump lasers.

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